Alu as the Driver of CNV Formation: A Computational Approach to Reveal NAHR

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Background

- Copy Number Variations (CNVs) are genomic alterations characterized by the gain or loss of DNA segments.
- CNVs alter the number of gene copies or regulatory elements, impacting gene function and expression.
- Understanding the mechanisms of CNV formation is essential for predicting CNV-mediated diseases. [1]
- As one of the rearrangement pathways, Non-allelic Homologous Recombination (NAHR) involves crossover events between homologous genome sequences on the same strand. [2][3]
- Especially, NAHR frequently occurs within repetitive elements such as Alu. [2][3]

Proximity and Spatial Association Test of Alu and CNVs

TABLE I. ENRICHMENT OF RETROTRANSPOSON OVERLAP COUNTS AT CNV BOUNDARIES

Retro trans posons	CNV loss			CNV gain		
	Observed statistics	Right-side p-value	Left-side p-value	Observed statistics	Right-side p-value	Left-side p-value
Alu	145.916	**0.002**[1]	1.000	-1.150	0.637	0.447
ERV1	13.668	0.0397	0.979	0.185	0.513	0.618
L1	-9.742	0.76530	0.257	5.516	0.182	0.852
L2	-21.106	0.990	0.006	2.482	0.263	0.832
MIR	-20.508	1.000	0.002	-1.926	0.815	0.318

1 Alu elements had significantly large overlap at CNV loss boundary regions

TABLE II. UNIQUE DISTANCE DISTRIBUTION FROM CNV BOUNDARIES TO RETROTRANSPOSON ELEMENTS

Family	CNV loss			CNV gain		
	Both boundary	Front boundary	Rear boundary	Both boundary	Front boundary	Rear boundary
LINE	0.196	0.004	0.320	0.044	0.409	0.085
LTR	0.571	0.343	0.763	0.571	0.343	0.763
SINE	**0.100**[2]	0.756	0.009	0.993	0.541	0.398

2 SINE family elements differ in overlapping distance distribution with significance (p-value < 0.1)

- If NAHR occurs through recombination between a pair of retrotransposon elements, homologous sequences would reside at CNV boundaries.
- We statistically tested the enrichment of retrotransposon overlaps at CNV boundaries using a permutation test, setting a null hypothesis that overlap counts of retrotransposons at CNV boundaries are similar to those at nonCNVs.
- Our analysis revealed that Alus exhibited a significantly high frequency of overlap at CNV loss boundaries within the human genome assembly (Table I).
- Further hypothesizing that distance distributions of Alus from CNVs would differ from those with nonCNVs, we statistically compared the distance distributions using the two-sample Z-test, t-test, and Wilcoxon signed-rank test.
- The results indicated that SINE elements exhibited a distinct distance distribution to their nearest CNV loss boundaries (Table II).
- These findings suggest that Alus play a pivotal role in mediating NAHR at CNV loss boundaries.

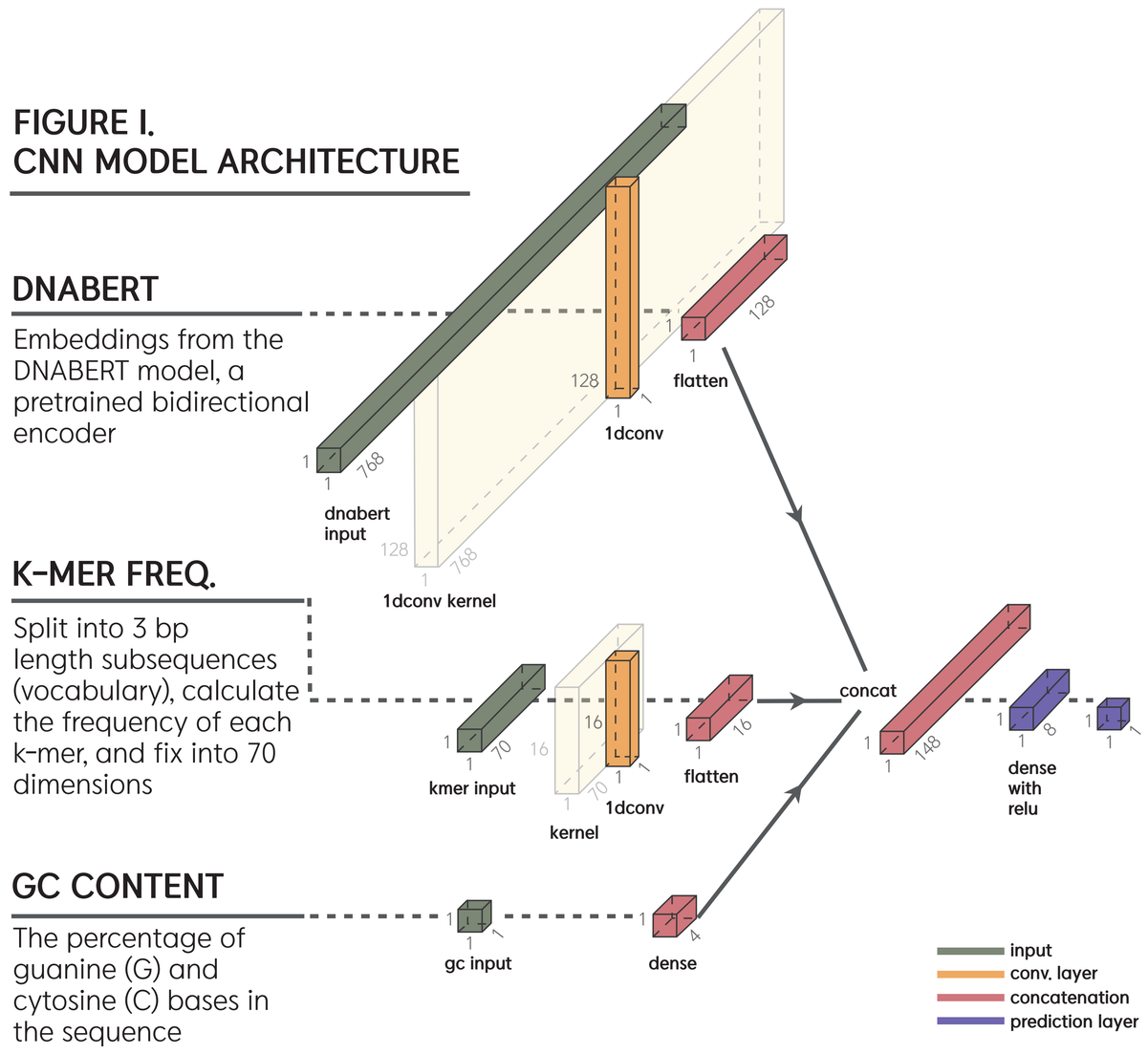
NAHR Prediction by Multi-Modal CNN

- We designed a multi-modal CNN model to predict whether an Alu sequence forms CNVs through NAHR, using only its genomic composition.
- The multi-modal CNN model comprises three distinct modalities to embed features separately: a Conv1D layer with kernel regularization for DNABERT, an identical Conv1D layer for K-mer frequency, and a dense layer for GC content (Figure I).
- The CNN model achieved an F1 score of 0.84, indicating that the genomic composition of Alus significantly contributes to CNV loss formation (Table III).
- This model not only effectively captures associations within genomic sequences, but also potentially functions as a detector for Alu-mediated CNV loss.

TABLE III. FEATURE SELECTION AND HYPER-PARAMETER TUNING FOR CNN MODEL

Dimension for each feature set			Metrics		
DNABERT	K-mer frequency	GC	Accuracy	F1 score	AUC
128	-	4	0.835	0.831	0.903
128	16	-	0.834	0.831	0.897
128	**16**	**4**	**0.846**	**0.840**	**0.930**

FIGURE I.
CNN MODEL ARCHITECTURE



Reference

[1] P Balachandran, I. A. Walawalkar, J. I. Flores, J. N. Dayton, P. A. Audano, C. R. Beck "Transposable element-mediated rearrangements are prevalent in human genomes," Nat Commun, vol. 13, April 2022, article. 7115.
[2] F. Zhang, W. Gu, M. E. Hurles, J. R. Lupski, "Copy number variation in human health, disease, and evolution," Annu Rev Genomics Hum Genet, vol. 10, Sep. 2009, pp. 451-81.
[3] X. Liao, W. Zhu, J. Zhou, H. Li, X. Xu, B. Zhang, X. Gao, "Repetitive DNA sequence detection and its role in the human genome," Commun Biol, vol. 6, Sept 2023, article. 954.

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